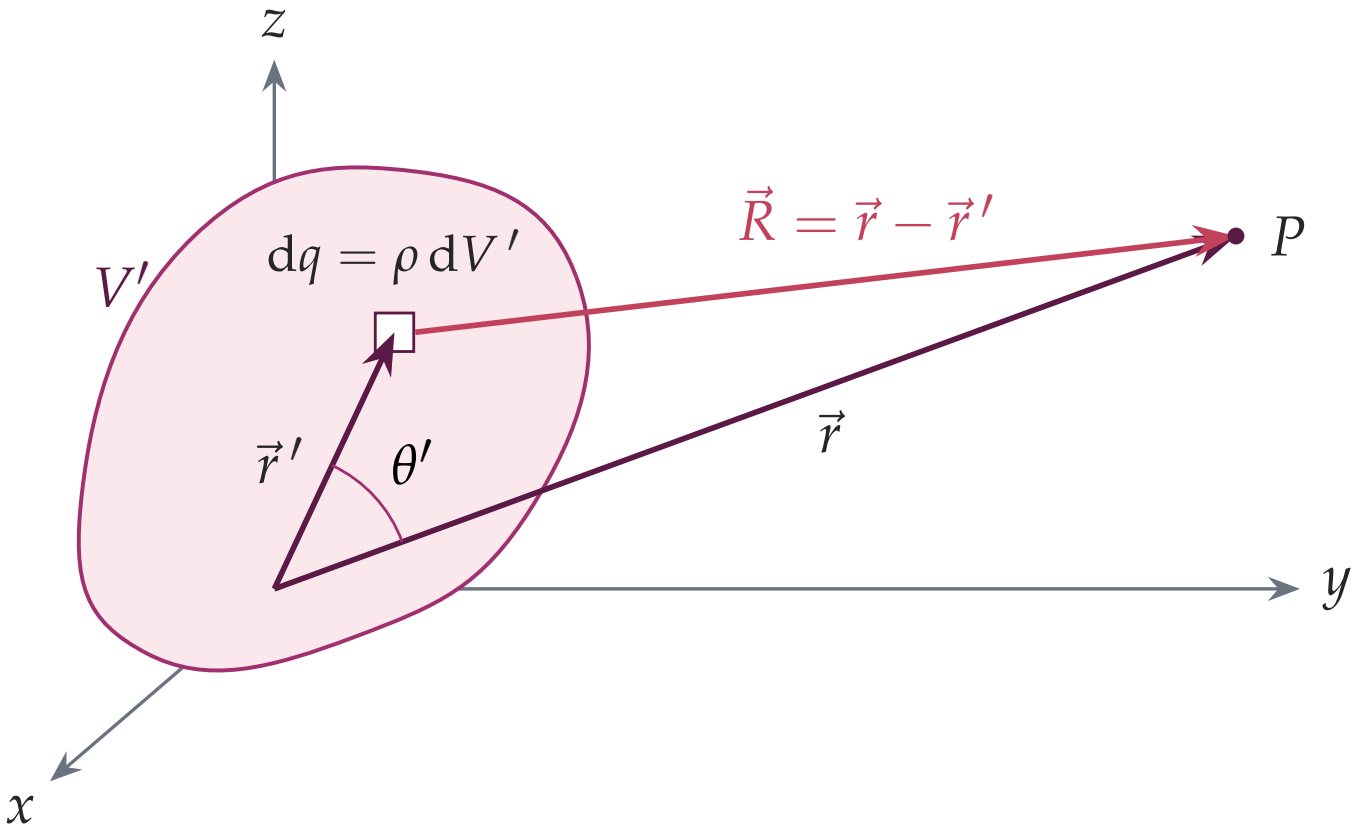

$$R^2 = r^2 + r'^2 - 2rr'\cos\theta'$$

$$\vec{r}\cdot\vec{r}\,' = rr'\cos\theta'$$